

MILIN ZHANG

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Education

Northeastern University

Ph.D Candidate in Computer Engineering
Advisor: Prof. Francesco Restuccia

Boston, MA
Apr 2026 (tentative)

Syracuse University

M.Sci in Electrical Engineering

Syracuse, NY
Dec 2021

University of Electronic Science and Technology of China

B.Eng in Electronic Engineering

Sichuan, China
June 2018

Research Interest

I am broadly interested in the intersection between artificial intelligence and wireless communication. My PhD research spans:

- **AI Security:** adversarial robustness, out-of-distribution detection, on-device model protection
- **Efficient AI in Distributed Systems:** edge computing, split computing, semantic communication
- **AI-Driven Wireless:** spectrum sensing, RF fingerprinting, integrated sensing and communication

Skills

Domain Expertise: Deep Learning, Statistical Learning, Digital Signal Processing, Wireless Communication, Convex Optimization, Information Theory

Coding: Python, C/C++, Matlab

Language: Chinese (Mandarin, Cantonese), English (TOEFL 108/120), Japanese (JLPT N1)

Publications

Journal

(*) denotes equal contribution

- Khandaker Foysal Haque, **Milin Zhang**, Francesca Meneghello, and Francesco Restuccia, “Si-FI: Learning the Beamforming Feedback for Simultaneous Multi-Subject Sensing.” *Computer Networks* (2025): 111907.
- **Milin Zhang**, Mohammad Abdi, Jonathan Ashdown, and Francesco Restuccia, “Adversarial Attacks to Latent Representations of Distributed Neural Networks in Split Computing.” *Computer Networks* (2025): 111755.
- **Milin Zhang***, Mohammad Abdi*, Venkat R. Dasari and Francesco Restuccia, “Semantic Edge Computing and Semantic Communications in 6G Networks: A Unifying Survey and Research Challenges.” in *Computer Networks* (2025): 111531.
- Khandaker Foysal Haque, **Milin Zhang**, Francesca Meneghello, and Francesco Restuccia, “BeamSense: Rethinking Wireless Sensing with MU-MIMO Wi-Fi Beamforming Feedback.” in *Computer Networks* (2025): 111020.

- Junyi Yang, Thomas Gourousis, Mengting Yan, Ruyi Ding, Ankit Mittal, **Milin Zhang**, Francesco Restuccia, Aatmesh Shrivastava, Yunsi Fei, and Marvin Onabajo, “A Low-Power Differential Temperature Sensor with Chopped Cascode Transistors and Switched-Capacitor Integration.” in *Electronics* 14.12 (2025): 2381.
- Ankit Mittal, **Milin Zhang**, Thomas Gourousis, Ziyue Zhang, Yunsi Fei, Marvin Onabajo, Francesco Restuccia, and Aatmesh Shrivastava, “Sub-6 GHz Energy Detection-based Fast On-Chip Analog Spectrum Sensing with Learning-driven Signal Classification.” in *IEEE Internet of Things Journal* 11.14 (2024): 25033-25046.

Conference

- **Milin Zhang**, Michael De Lucia, Jonathan Ashdown, Nathaniel D. Bastian, Ananthram Swami, and Francesco Restuccia, “NI-Diff: Zero-Day and Adversarial Network Intrusion Detection with Diffusion Models.” in *Proc. of IEEE Military Communications Conference (MILCOM)*, 2025
- **Milin Zhang**, Mohammad Abdi, Shahriar Rifat, and Francesco Restuccia, “Resilience of Entropy Model in Distributed Neural Networks.” in *Proc. of the 18th European Conference on Computer Vision (ECCV)*, 2024.
- Daniel Uvaydov*, **Milin Zhang***, Clifton Paul Robinson, Salvatore D’Oro, Tommaso Melodia and Francesco Restuccia, “Stitching the Spectrum: Semantic Spectrum Segmentation with Wideband Signal Stitching.” in *Proc. of IEEE Conference on Computer Communications (INFOCOM)*, 2024.
- **Milin Zhang**, Michael De Lucia, Ananthram Swami, Jonathan Ashdown, Kurt Turck and Francesco Restuccia, “HyperAdv: Dynamic Defense Against Adversarial Radio Frequency Machine Learning Systems.” in *Proc. of IEEE Military Communications Conference (MILCOM)*, 2024
- Khandaker Foysal Haque, **Milin Zhang**, Francesco Restuccia, “SiMWiSense: Simultaneous Multi-Subject Activity Classification Through Wi-Fi Signals.” in *Proc. of the IEEE 24th International Symposium on World of Wireless, Mobile and Multimedia Networks (WoWMoM)*, 2023.
- Thomas Gourousis, Ziyue Zhang, Mengting Yan, **Milin Zhang**, Ankit Mittal, Aatmesh Shrivastava, Francesco Restuccia, Yunsi Fei, and Marvin Onabajo, “Identification of Stealthy Hardware Trojans through On-Chip Temperature Sensing and an Autoencoder-Based Machine Learning Algorithm.” in *Proc. of the IEEE 66th International Midwest Symposium on Circuits and Systems (MWSCAS)*, 2023.

In Preparation

- **Milin Zhang***, Tanzil Hassan*, Mohammad Abdi, Venkat R. Dasari and Francesco Restuccia, “MD2I: Multi-Device Model-Distributed Neural Network Inference.” Submitted to the 46th IEEE International Conference on Distributed Computing Systems.
- **Milin Zhang** and Francesco Restuccia, “T-MUX: Securing Neural Networks with Task Multiplexing.” Submitted to the 42nd Conference on Uncertainty in Artificial Intelligence.

Preprint

- Ildi Alla, **Milin Zhang**, Jonathan Ashdown, Valeria Loscri and Francesco Restuccia, “Finding a Needle in a (Spectrum) Haystack: Passwordless Wireless Authentication Through Multi-Band Multi-Device Radio Fingerprinting.” (2025).
- Sayyed Sazzad, Shahriar Rifat, **Milin Zhang**, Ananthram Swami, Michael De Lucia, Nathaniel D. Bastian, and Francesco Restuccia, “Out-of-Distribution Detection in Computer Vision: A Comprehensive Survey and Research Challenges.” (2025).
- Sayyed Sazzad*, **Milin Zhang***, Shahriar Rifat*, Ananthram Swami, Michael De Lucia, and Francesco Restuccia, “Resilience and Security of Deep Neural Networks Against Intentional and Unintentional Perturbations: Survey and Research Challenges.” *arXiv preprint arXiv:2408.00193* (2024).

Patents

- Daniel Uvaydov, **Milin Zhang**, Salvatore D'Oro, Tommaso Melodia, Francesco Restuccia, and Clifton Paul Robinson, "Methods for Real-Time Wideband RF Waveform and Emission Classification." U.S. Patent Application 18/620,310, filed October 3, 2024.
- Francesco Restuccia, Khandaker Foysal Haque, and **Milin Zhang**, "Simultaneous Multi-Subject Activity Classification Through Wi-Fi Signals." U.S. Patent Application 18/489,570, filed June 6, 2024.
- Francesco Restuccia, Khandaker Foysal Haque, and **Milin Zhang**, "Method and Apparatus for Wi-Fi Sensing Through MU-MIMO Beamforming Feedback Learning." WO 2024/049970, filed August 31, 2023.

Research Software

I am committed to reproducibility and open science, releasing documented implementations to enable the research community to validate and build upon my work.

- AdvLatent: Adversarial attack framework for distributed neural networks
<https://github.com/Restuccia-Group/AdvLatent>
- EntropyR: Robustness evaluation framework for entropy models in distributed neural networks
<https://github.com/Restuccia-Group/EntropyR>
- HyperAdv: Dynamic defense toolkit against adversarial RF machine learning attacks
<https://github.com/Restuccia-Group/HyperAdv>
- Spectrum Stitching: Wideband signal processing and segmentation framework for spectrum sensing
https://github.com/uvaydovd/spectrum_sensing_stitching

Service

Reviewer

- Journals: IEEE J-STSP, TCOM, TCCN, TWC, TMC, TDSC, and Elsevier COMNET.
- Conferences: ICC, GLOBCOM, SECON, WoWMoM, MILCOM, AAAI, ECCV

Teaching Assistant

- EECE 5643-Simulation and Performance Evaluation, Spring 2026